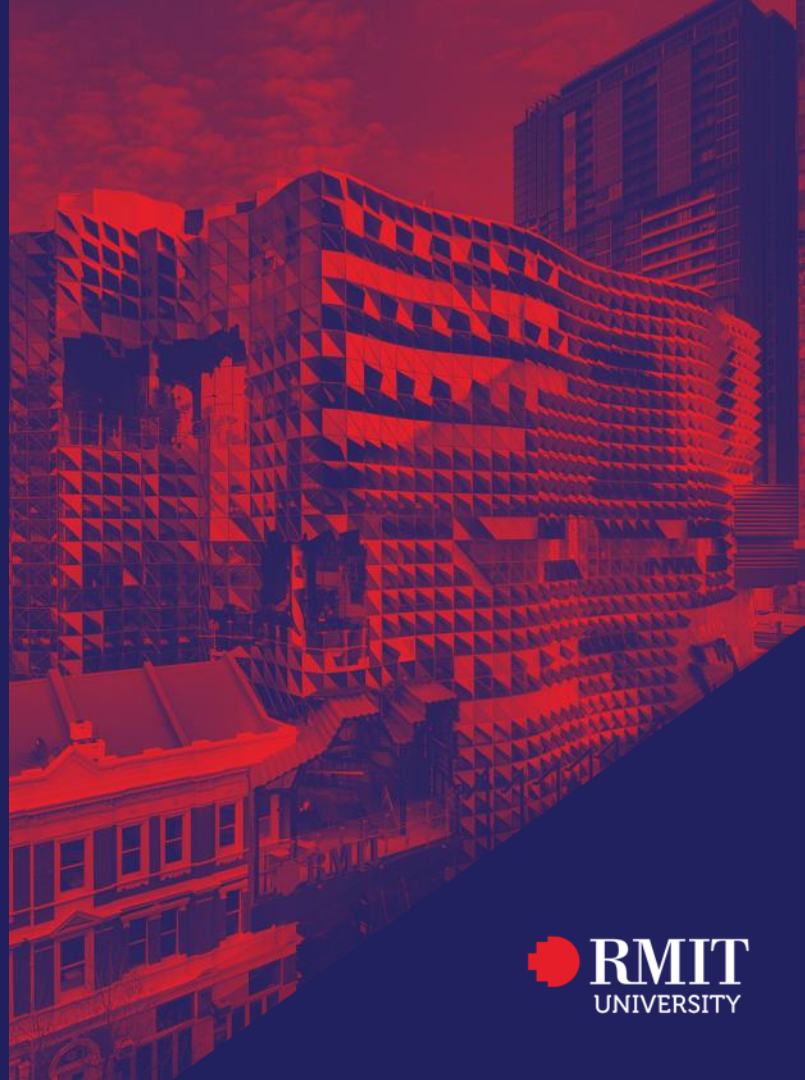
# ISYS 1108 / 3474 Software Engineering Project Management

# Week 5: Workshop

Dr Maria Spichkova

Senior Lecturer in Software Engineering

What's next...



RMIT UNIVERSITY

# Acknowledgement of Country

*RMIT University acknowledges the people of the Woi wurrung and Boon wurrung language groups of the eastern Kulin Nation on whose unceded lands we conduct the business of the University.*

*RMIT University respectfully acknowledges their Ancestors and Elders, past and present.*

*RMIT also acknowledges the Traditional Custodians and their Ancestors of the lands and waters across Australia where we conduct our business.*

# Our plan for this session

- 😊 Sprint Planning Notes
- 😊 Sprint Retro Notes
- 😊 Trello Retro board
- 😊 Discussion on the results of SE Challenge 1
- 😊 Discussion on Team-based assignment

## Activity 1: Sprint Planning Notes and Sprint Retro Notes – general discussion

Duration: approx. 10 minutes

Discuss with the tutor the following questions:

- Who should create these notes and when?
- How much time this work should take?
- How many pages you typically need to write?

# Sprint Planning Notes
# Sprint Retro Notes

**Who should create them and when:**
The notes have to be written by the Scrum Master **during** the corresponding meeting.

**How much time this work should take:** No extra time on top of the meeting time, you have to create them during the meeting (scribe the core points of the discussion).

**How many pages you typically need to write:**
The notes should be short, max 2 pages.
Write short simple sentences, avoid having long paragraphs. Keep only necessary details. Proof-read!

**Retro notes should match to your Planning Notes!**

# Sprint Planning Notes

Canvas ➔ Modules ➔ Templates, Tools, Team-allocation ➔ Useful Templates

Short notes – max. 2 pages

### 1. Goal

What is the aim that you are trying to achieve by running this sprint?

### 2. Duration of the sprint

2 weeks

### 3. What is the team's vision for this sprint?

Which items of the product backlog will be committed to sprint backlog and why?

What will the potentially shippable product look like in the end?

What features will it have in its working form?

### 4. Estimation in story points

Team to estimate story points for each of the items. Provide some sort of justification.

# Sprint Retro Notes

Canvas ➔ Modules ➔ Templates, Tools, Team-allocation ➔ Useful Templates

**1. Things That Went Well**

What went well? What the team is happy about?

**2. Things That Could Have Gone Better**

What could have gone better? What the team could improve?

**3. Things That Surprised Us**

What wasn't expected?

**4. Lessons Learned**

What you learned from the above points?

**5. Final Thoughts**

Things to Keep

Things to Change

Short notes – max. 2 pages

# Activity 2: Sprint Planning Notes – Example 1

## SprintPlanningNotes_Sprint1_OPM.pdf

Duration: approx. 15 minutes

SPRINT PLANNING NOTES — DATE OF THE MEETING: 27 FEBRUARY 2023

**Project: OPM (Online Patient Management)** **Team: X1**

| | Attended | Absent |
|---|---|---|
| Scrum Master | Severus Snape | |
| Product Owner | Dr Strange | |
| Development Team | Harry Potter, Ron Weasley, Hermione Granger, Neville Longbottom, Luna Lovegood, Draco Malfoy | |
| Customers (optional) | | |

| Sprint duration | Sprint # | Start Day | End Day | Total Working Days | Expected capacity | Expected Velocity |
|---|---|---|---|---|---|---|
| 2 weeks | 1 | 27 February 2023 | 10 March 2023 | 10 | 100% | 27 story points |

1. Team's goal (vision) for this sprint

The aim of this sprint is to create the first shippable product increment that covers the basic functionality on patient management.

The following features will be implemented as the basic functionality on patient management:

1. Login features of all type (PBI 3)
2. Patient registration (PBI 1)
3. Patient search (PBI 24)
4. Patient record modification (PBI 8)
5. Adding information to patient record (PBI 12)

These features will allow to demonstrate to the end-users how this part of the tool might be applied, and the Product Owner requested to implement the above functionalities.

2. Scope of the sprint: Estimation

***PBI 3:*** *As an authorised user, I want to log into system so that I can gain authorisation.*
3 story points – the implementation of this functionality will be easy, but we will need to create the corresponding data base tables first, which will also take some time (we are going to start from this PBI).

SPRINT PLANNING NOTES — DATE OF THE MEETING: 27 FEBRUARY 2023

***PBI 1:*** *As an admin, I want to register new patients so that there is a record of them in the system.*
13 story points – we will need to implement the checks for input fields to avoid incorrect registration.

***PBI 24:*** *As a doctor / nurse I want to search for a patient records so that I can view and update patients' data*
5 story points – we will implement three types of search (by the last name, by the phone number, by Medicare number) but all three types are very similar in implementation and the query is simple. The complexity of this story is in the visualisation of the retrieved data (this functionality should be covered by PBI 24 as well)

***PBI 8:*** *As an admin/ doctor / nurse I want to edit patient information so that the patient record is up-to-date.*
3 story points – this will be easy to implement, as we will reuse many functions created for PBI 1 (PBI 1 should be implemented first).

***PBI 12:*** *As a doctor/ nurse I want to add notes to a patient's record, so that I can inform other staff of any extra details about the patient, outside of the general patient information.*
3 story points – we will implement new checks for input fields to avoid incorrect data input (might also require some cross checks with previous entries for this patient), this will take time.

These notes are correct.
Discuss them within your team and then with the tutor, section by section.

SPRINT PLANNING NOTES DATE OF THE MEETING 27 FEBRUARY 2023

**Project: OPM (Online Patient Management)** **Team: X1**

| | Attended | Absent |
|---|---|---|
| Scrum Master | Severus Snape | |
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| Customers (optional) | | |

| Sprint duration | Sprint # | Start Day | End Day | Total Working Days | Expected capacity | Expected Velocity |
|---|---|---|---|---|---|---|
| 2 weeks | 1 | 27 February 2023 | 10 March 2023 | 10 | 100% | 27 story points |

Correct! The meeting as at the very first day of the sprint.

# Let’s look at another correct version:

SPRINT PLANNING NOTES DATE OF THE MEETING: 6 MARCH 2023

**Project: OPM (Online Patient Management)** **Team: X1**

| | Attended | Absent |
|---|---|---|
| Scrum Master | Severus Snape | |
| Product Owner | Dr Strange | |
| Development Team | Harry Potter, Hermione Granger, Neville Longbottom, Luna Lovegood, Draco Malfoy | Ron Weasley |
| Customers (optional) | | |

| Sprint duration | Sprint # | Start Day | End Day | Total Working Days | Expected capacity | Expected Velocity |
|---|---|---|---|---|---|---|
| 2 weeks | 1 | 6 March 2023 | 17 March 2023 | 9 | 85% | 27 story points |

*Monday 13 March is a public holiday in Victoria (Labour Day). Ron is on sick leave and expects to return only next sprint.*

## 1. Team's goal (vision) for this sprint

The aim of this sprint is to create the first shippable product increment that covers the basic functionality on patient management.

The following features will be implemented as the basic functionality on patient management:

1. Login features of all type (PBI 3)
2. Patient registration (PBI 1)
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These features will allow to demonstrate to the end-users how this part of the tool might be applied, and the Product Owner requested to implement the above functionalities.

## 2. Scope of the sprint: Estimation

***PBI 3:*** *As an authorised user, I want to log into system so that I can gain authorisation.*

3 story points – the implementation of this functionality will be easy, but we will need to create the corresponding data base tables first, which will also take some time (we are going to start from this PBI).

***PBI 1:*** *As an admin, I want to register new patients so that there is a record of them in the system.*

13 story points – we will need to implement the checks for input fields to avoid incorrect registration.

***PBI 24:*** *As a doctor / nurse I want to search for a patient records so that I can view and update patients' data*

5 story points – we will implement three types of search (by the last name, by the phone number, by Medicare number) but all three types are very similar in implementation and the query is simple. The complexity of this story is in the visualisation of the retrieved data (this functionality should be covered by PBI 24 as well)

***PBI 8:*** *As an admin/ doctor / nurse I want to edit patient information so that the patient record is up-to-date.*

3 story points – this will be easy to implement, as we will reuse many functions created for PBI 1 (PBI 1 should be implemented first).

***PBI 12:*** *As a doctor/ nurse I want to add notes to a patient's record, so that I can inform other staff of any extra details about the patient, outside of the general patient information.*

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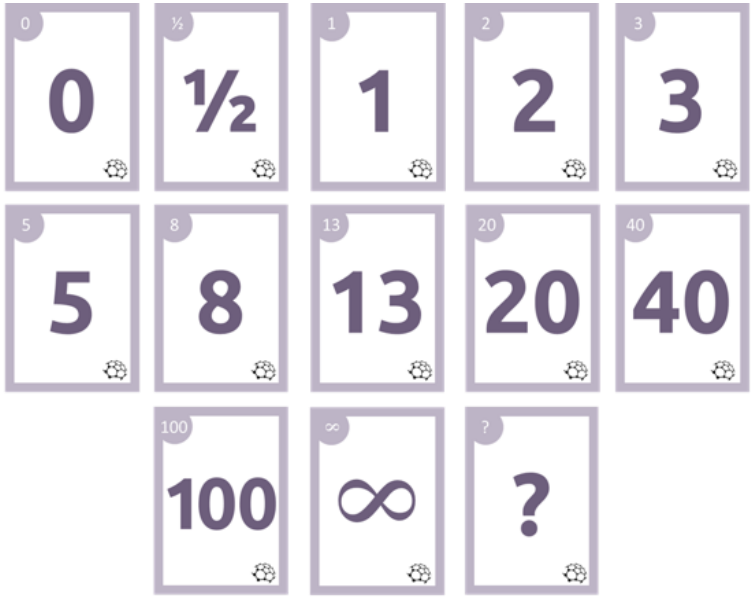
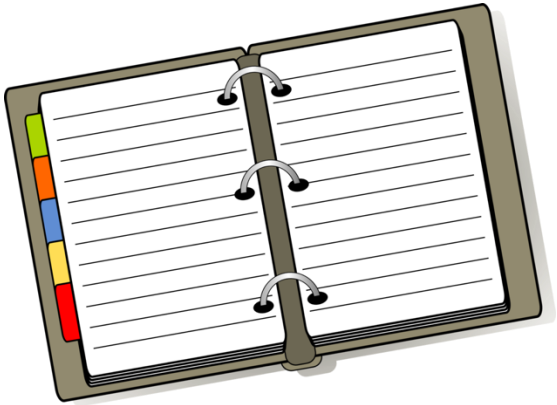


# Activity 3: Sprint Planning Notes – Example 2

Duration: approx. 15 minutes

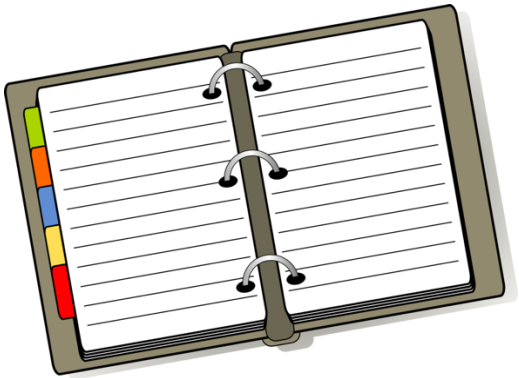
Check the document
SprintPlanningNotes_Sprint1_withMistakes.pdf

What are the mistakes in these notes?
Discuss this within your team and then with your tutor.



# Let’s discuss SprintPlanningNotes_Sprint1_withMistakes.pdf

- ✗ Scrum Master and Product Owner are the same person. Either this is a typo that should be fixed, or the issue is in role allocation. Scrum Master and Product Owner roles should not be allocated to the same person.
- ✗ This version of the goal makes no sense – you cannot have high-level good-praxis rules as a goal, the goal should be specific for the sprint and focusing on the functionality
- ✗ Duration of a sprint cannot be 1.5 weeks – we need to have 1 or 2 week(s) duration, same for all sprint.
- ✗ Planning meeting should be the very first activity in the sprint, but the sprint starts on 27 February and the meeting was on 1 March. This is incorrect.
- ✗ Justification for estimates are missing.

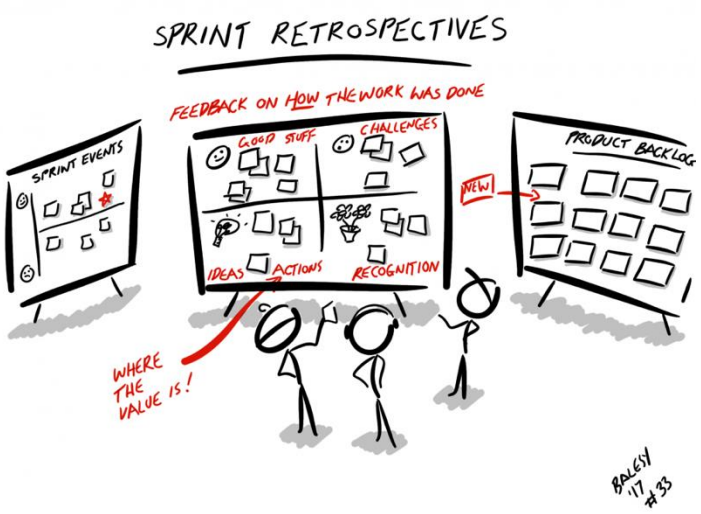


Let’s compare it with the correct version…

# Activity 4: Sprint Retro Notes – Example 1

## SprintRetroNotes_Sprint1_OPM.pdf

Duration: approx. 15 minutes




These notes are correct.
Discuss them with tutor section by section.

SPRINT RETRO NOTES — DATE OF THE MEETING: 10 MARCH 2022

**Project:** OPM (Online Patient Management) **Team:** X1

| | Attended | Absent |
|---|---|---|
| Scrum Master | Severus Snape | |
| Development Team | Harry Potter, Ron Weasley, Hermione Granger, Neville Longbottom, Luna Lovegood, Draco Malfoy | |

Sprint #: 1

1. What went well?

*We completed all PBIs planned for this sprint and were able to implement two extra PBI from the top of the product backlog.*

*We were able to use Selenium for automated testing – this was our first time using it for a real project!*

2. What went not so well?

*Estimation of the team's capacity: we underestimated the team's capacity.*

*Demo: The demo at the Sprint review meeting was not so good, the customers understood only at the very end of the meeting what exactly is now fully completed. Also, the battery of the laptop on which we were planning to present became empty at the middle of demo.*

3. Lessons learned

*Estimates: We can do more than 15 story points per sprint.*

*Demo: We need to have a team-internal dry-run of the demo before the Sprint review meeting. We need to check whether the demo-laptop is charged and that we have a back-up laptop also prepared for demo.*

*We haven't done automated testing before, and we were surprised how much time we saved on regression testing.*

***Things to Keep:*** *Automated testing with Selenium*

***Things to Change:*** *We will plan for 25 story points as team's velocity, plan team-internal dry-run of the demo, backups for demo.*

SPRINT RETRO NOTES DATE OF THE MEETING: 10 MARCH 2022

**Project: OPM (Online Patient Management)** **Team: X1**

| | Attended | Absent |
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***Things to Keep:*** *Automated testing with Selenium*

***Things to Change:*** *We will plan for 25 story points as team's velocity, plan team-internal dry-run of the demo, backups for demo.*

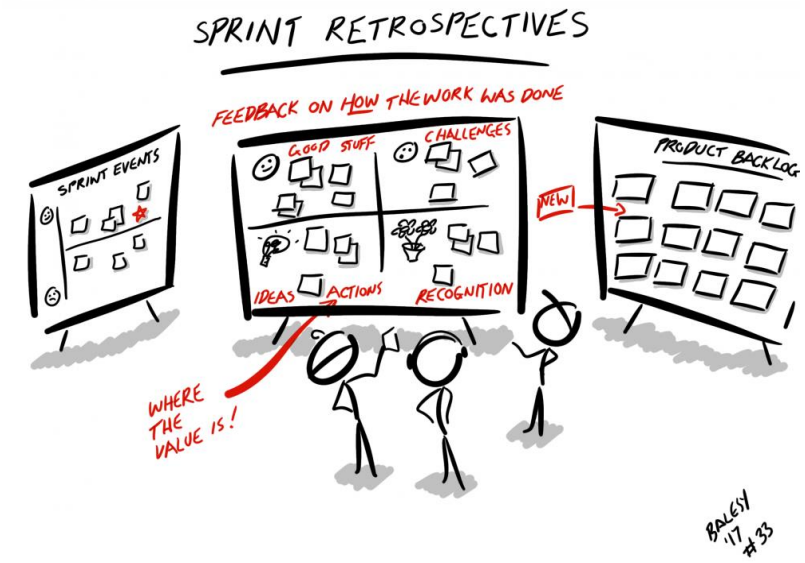
# Activity 5: Sprint Retro Notes – Example 2

Duration: approx. 20 minutes

Check the document SprintRetroNotes_Sprint1_withMistakes.pdf

What are the mistakes in these notes? Discuss this within your team and then with your tutor.

Let's compare these notes with the correct version in Example 1…



# Let’s discuss SprintRetroNotes_Sprint1_withMistakes.pdf (1)

- ✗ Date: either it’s typo, or the meeting wasn’t held at the very end of the sprint, which is incorrect.
- ✗ Product Owner and customers should not attend the retro meeting, it should be a team-internal meeting (only the team and the Scrum Master should attend)
- ✗ The tone of this version is aggressive, focus is on blaming rather than on improving the situation

# Let’s discuss SprintRetroNotes_Sprint1_withMistakes.pdf (2)

✗ This version of “Things that went well (What went well)” is unacceptable:

- The first paragraph – the content is correct, but this text doesn’t belong to the notes. The notes should be short and focused on actual points discussed at the meeting.
- The second and the third paragraphs: aggressive/unfriendly tone, details out of scope.

# Let’s discuss SprintRetroNotes_Sprint1_withMistakes.pdf (3)

✗ This version of “Things that could have gone better (What went no so well)” is unacceptable:

- The first paragraph – the same issue as with the first section. The content is correct, but this text doesn’t belong to the notes. The notes should be short and focused on actual points discussed at the meeting.
- The second paragraph: too wordy, too many introductory sentences/ phrases, so many unnecessary details that it’s even hard to identify what was actually the point to improve.
- The third paragraph: too many unnecessary details, aggressive tone, blame shifting

# Let’s discuss SprintRetroNotes_Sprint1_withMistakes.pdf (4)

✗ This version of “Lessons learned” is unacceptable:

- The first and second paragraph – the same issue as with the first section. The content is correct, but this text doesn’t belong to the notes. The notes should be short and focused on actual points discussed at the meeting.
- The third paragraph: too wordy, too many unnecessary details.
- The fourth paragraph: out of scope. Even when the cloak of levitation might be a real surprise, it’s out of scope here.
- This version of “Final thoughts” is unacceptable – neither “things to keep” nor “things to change” make any sense in this version.

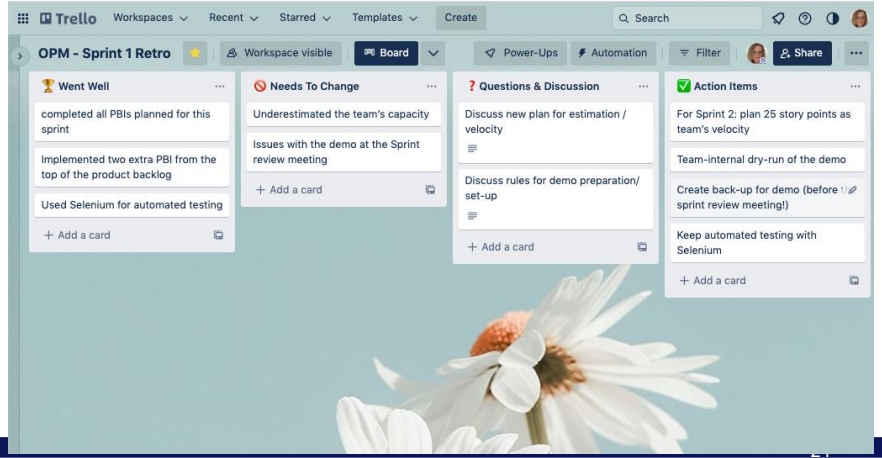
# Activity 6: Sprint Retro Board in Trello

**Duration: approx. 15 minutes**

There is an alternative to retro notes document – a retro board, it can be created in Trello.

Let's compare the document SprintRetroNotes_Sprint1_OPM.pdf
With the corresponding Trello RETRO board

Note: A Trello RETRO board is NOT the same as a Trello board for a Scrum project (Trello Scrum board)




SPRINT RETRO NOTES — DATE OF THE MEETING: 10 MARCH 2022

**Project:** OPM (Online Patient Management) — **Team:** X1

| | Attended | Absent |
|---|---|---|
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Sprint #: 1

1. What went well?

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2. What went not so well?

*Estimation of the team's capacity: we underestimated the team's capacity.*

*Demo: The demo at the Sprint review meeting was not so good, the customers understood only at the very end of the meeting what exactly is now fully completed. Also, the battery of the laptop on which we were planning to present became empty at the middle of demo.*

3. Lessons learned

*Estimates: We can do more than 15 story points per sprint.*

*Demo: We need to have a team-internal dry-run of the demo before the Sprint review meeting. We need to check whether the demo-laptop is charged and that we have a back-up laptop also prepared for demo.*

*We haven't done automated testing before, and we were surprised how much time we saved on regression testing.*

***Things to Keep:*** *Automated testing with Selenium*

***Things to Change:*** *We will plan for 25 story points as team's velocity, plan team-internal dry-run of the demo, backups for demo.*

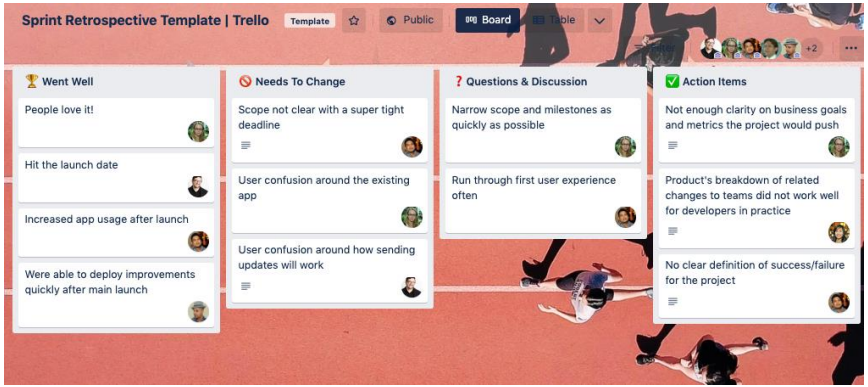
# Sprint Retro Board in Trello: General points

**Easier to read:**

- visual,
- potentially less cognitive load,
- Push you to be short/precise in wording

**Missing information:**

- Attendance
- Dates




https://trello.com/b/hkaQsLWx/sprint-retrospective-template-trello

Trello | Workspaces | Recent | Starred | Templates | Create | Search

## OPM - Sprint 1 Retro

Workspace visible | Board | Power-Ups | Automation | Filter | Share

### 🏆 Went Well

- completed all PBIs planned for this sprint
- Implemented two extra PBI from the top of the product backlog
- Used Selenium for automated testing

+ Add a card

### 🚫 Needs To Change

- Underestimated the team's capacity
- Issues with the demo at the Sprint review meeting

+ Add a card

### ❓ Questions & Discussion

- Discuss new plan for estimation / velocity
- Discuss rules for demo preparation/ set-up

+ Add a card

### ✅ Action Items

- For Sprint 2: plan 25 story points as team's velocity
- Team-internal dry-run of the demo
- Create back-up for demo (before the sprint review meeting!)
- Keep automated testing with Selenium

+ Add a card

Trello | Workspaces | Recent | Starred | Templates | Create | Search

OPM - Sprint 1 Retro | Workspace visible | Board | Power-Ups | Automation | Filter | Share

**Went Well**

- completed all PBIs planned for this sprint
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- Used Selenium for automated testing

+ Add a card

**Needs To Change**

- Underestimated the team's capacity
- Issues with the demo at the Sprint review meeting

+ Add a card

**Questions & Discussion**

- Discuss new plan for estimation / velocity
- Discuss rules for demo preparation/

**Action Items**

- For Sprint 2: plan 25 story points as team's velocity
- Team-internal dry-run of the demo

1. What went well?

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2. What went not so well?

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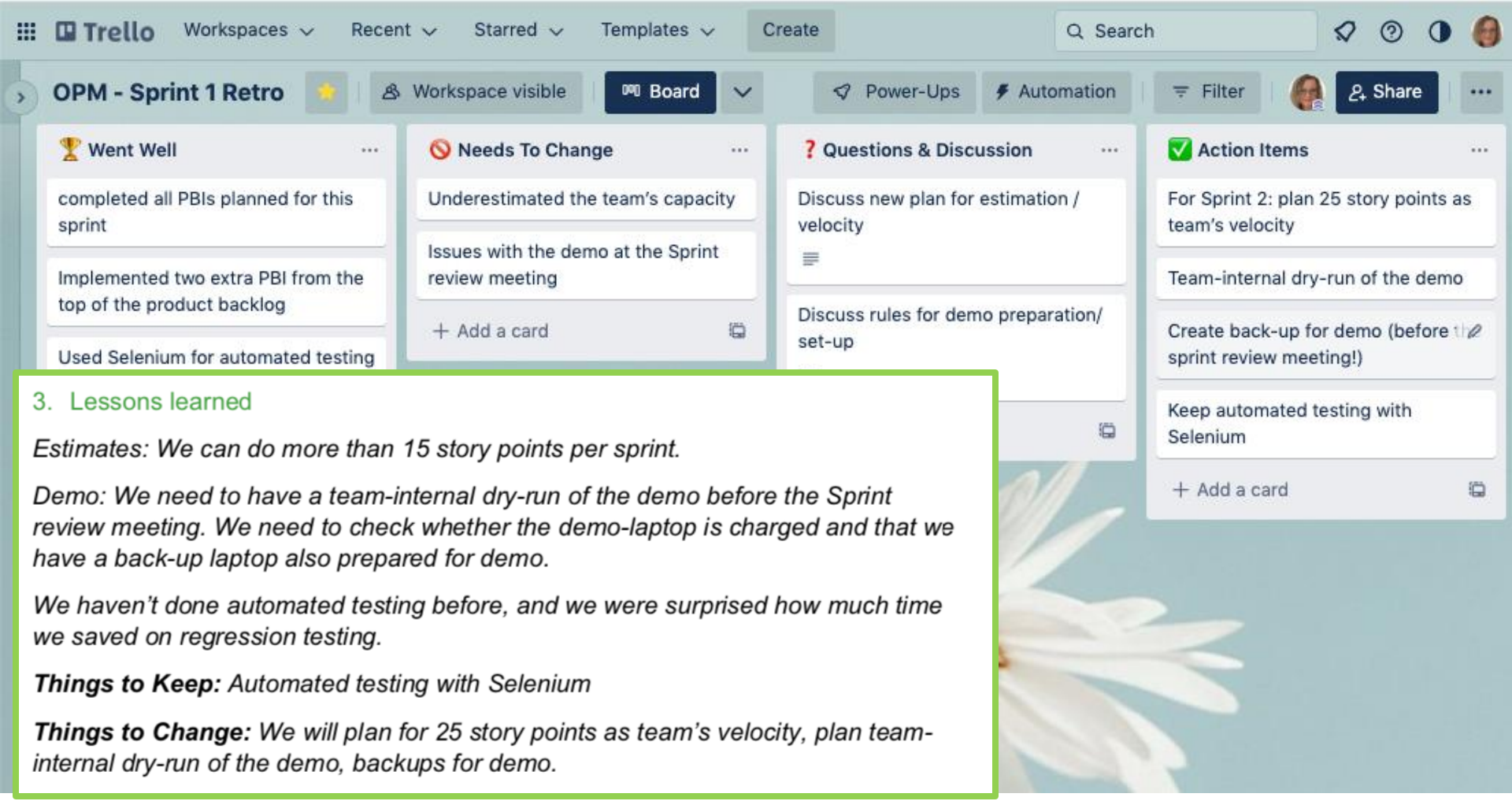


## 3. Lessons learned

*Estimates: We can do more than 15 story points per sprint.*

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+ Add a card

### Discuss new plan for estimation / velocity

in list ❓ Questions & Discussion

Notifications

Watch

**Description** Edit

*We can do more than 15 story points per sprint.*

**Activity** Hide Details

Write a comment...

Add to card

Members

Labels

Checklist

Dates

Attachment

Cover

Custom Fields

Trello | Workspaces | Recent | Starred | Templates | Create | Search

# OPM - Sprint 1 Retro

Workspace visible | Board | Power-Ups | Automation | Filter | Share

## Went Well

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## Needs To Change

- Underestimated the team's capacity
- Issues with the demo at the Sprint review meeting

+ Add a card

## Questions & Discussion

- Discuss new plan for estimation / velocity
- Discuss rules for demo preparation/ set-up

+ Add a card

## Action Items

- For Sprint 2: plan 25 story points as team's velocity
- Team-internal dry-run of the demo
- Create back-up for demo (before the sprint review meeting!)
- Keep automated testing with Selenium

+ Add a card

### Discuss rules for demo preparation/ set-up

in list ❓ Questions & Discussion

Notifications

Watch

**Description** Edit

*We need to have a team-internal dry-run of the demo before the Sprint review meeting. We need to check whether the demo-laptop is charged and that we have a back-up laptop also prepared for demo*

**Activity** Hide Details

Write a comment...

Add to card

Members

Labels

Checklist

Dates

Attachme

Cover

Custom F

# Activity 7: Discussion on the results of SE Challenge 1

Duration: approx. 10 minutes

Have you checked your results of SE Challenge 1?
Do you have any questions on your results?

Don’t hesitate to ask tutor at the workshop if you have any questions.

# Activity 8: Discussion on Team-based assignment

**Step 1: Analyse the scenario, which was inspired by a real-life problem:**
Assume that a **Scrum team** is working on a software development project. Sprint 1 of the project starts on 15 September. The project will be planned for 6 weekly sprints.
Assume in this scenario that

- The Scrum team consist of all members of your team.
- Dr Maria Spichkova is the Product Owner.
- Your workshop tutor is the Scrum Master.
- Osheen Walawedura is the customer who requested access to the information radiator.

The goal of the project is to develop a software system for a small local company producing handmade scarves, which can be picked up from their small shop or delivered. Currently, they use a simple Excel spreadsheet to plan their orders, but as their business is growing, they would like to automate the ordering process and have a cashless payment only. **For the details of the required features of the system, please read the *Notes made during the initial requirements elicitation interviews with the stakeholders***. Like in a real project, these notes are not fully structured and have to be analysed and organised by you: you will need to analyse the notes, identify the details of the functionalities to implement, including their priorities for the project.

**Step 2:** Based on your analysis of the above notes, collaboratively create the following artefacts ***during the Week 6 and 7 workshop sessions***: …

Duration:
approx. 10 minutes

Do you know who is in your team?
Do you have any questions to the tutor?
Have you read the assignment specification?

Don't hesitate to ask tutor at the workshop if you have any questions!

# Please check the marking criteria (below you can see the first 2 criteria)

| Criteria | Ratings | | | | Pts |
|---|---|---|---|---|---|
| Trello/Jira board layout | 3 pts<br>**Full marks**<br>The board has correct format/layout as expected for Trello/Jira approach, including the presentation on PBI IDs, using labels, etc. | 2 pts<br>**Good**<br>There are some mistakes, but the solution is generally reasonable. See feedback from the marker. | 1 pts<br>**Needs improvement**<br>The solution contains many critical mistakes. See feedback from the marker. | 0 pts<br>**No marks**<br>No link to the board submitted, or the board is completely incorrect, or you haven't contributed to the creation of this artefact. | 3 pts |
| Completeness of the product backlog | 3 pts<br>**Full Marks**<br>The set of cards presenting all PBIs (all cards on the boards) covers all functionalities requested by the client, none of the PBIs is out of scope, all items are correctly sorted by priorities. The spreadsheet version of the product backlog matches the board. | 2 pts<br>**Good**<br>There are some mistakes, but the solution is generally reasonable. See feedback from the marker. | 1 pts<br>**Needs improvement**<br>The solution contains many critical mistakes or the spreadsheet-version of the product backlog is missing. See feedback from the marker. | 0 pts<br>**Incorrect**<br>No link to the board submitted, or the product backlog is completely incorrect/missing, or you haven't contributed to the creation of the product backlog. | 3 pts |